

TEACHER'S NAME: _____

STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

YEAR 12

YEAR 12 HALF YEARLY EXAMINATION

2008

**MATHEMATICS
EXTENSION 2**

*Time allowed - Three hours
(Plus five minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Begin each Question on a fresh page.
- All necessary working should be shown. Marks shown are a guide only
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

Question 1**MARKS**

a) If $z = \frac{2+i}{2-i}$, find

(i) z in the form $a+ib$ (where a and b are real)

1

(ii) $|z|$

1

(iii) $\text{Arg } z$, correct to the nearest degree

1

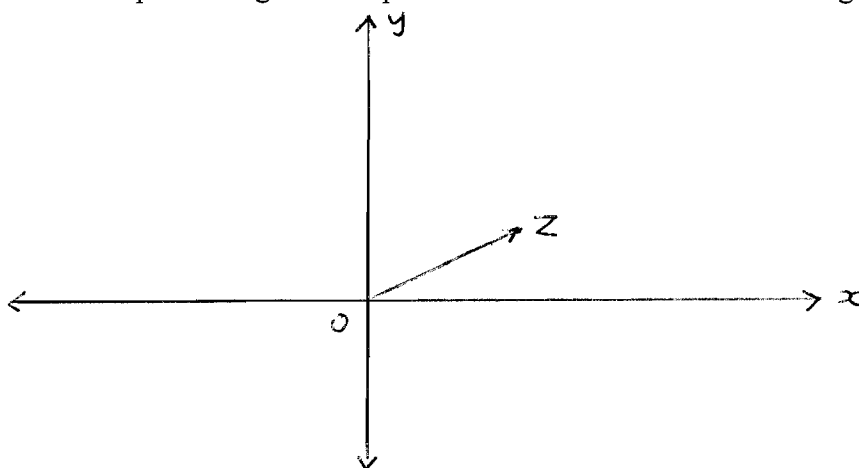
(iv) $\bar{z}$

1

(v) $\text{Im}(iz)$

1

b) The point and vector representing the complex number z are shown on the Argand diagram.



Copy the diagram, and plot and label the points representing $\bar{z}$, $2z$, iz and $(2+i)z$

4

c) Find the two square roots of $4-3i$

3

d) If $w = \sqrt{3} + i$, express w in modulus-argument form and hence evaluate w^6

3**Question 2**

a) Sketch on an Argand diagram:

(i) $|z-1| = |z+i|$

1

(ii) the region where $|z| \leq 1$ and $0 \leq \arg z \leq \frac{\pi}{3}$

3

Question 2 (cont)

b) Sketch the locus of z if $|z - (1 + i)| = 2$. For this locus, what is the maximum value of:

(i) $|z|$

(ii) $\arg z$

3

c) Use De Moivre's theorem and the expansion of $(\cos \theta + i \sin \theta)^3$ to obtain an expression for $\cos^3 \theta$, and hence find $\int \cos^3 \theta \cdot d\theta$

5

d) $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$.

Prove that $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

3

Question 3

(a) If $x^2 - y^2 = 0$, show that $\frac{dy}{dx} = \frac{x}{y}$ and express $\frac{d^2y}{dx^2}$ in its simplest form.

3

b) (i) Sketch the ellipse $\frac{x^2}{16} + y^2 = 1$, showing the intercepts, foci and directrices

4

(ii) Write down the parametric equations of this ellipse

2

(iii) Find the equation of the tangent to this ellipse at the point on it where $x=2$ and $y>0$.

3

c) Sketch the locus of the complex number $z=x+iy$ if $|z - 3| + |z + 3| = 10$.

Find the Cartesian equation of this locus.

3

Question 4

a) $P(ct, \frac{c}{t})$ and $Q(\frac{ct}{3}, \frac{3c}{t})$ are two points on the rectangular hyperbola $xy = c^2$ ($t>0$).

(i) Sketch the hyperbola and draw and label the chord PQ.

1

(ii) Find the coordinates of M, the midpoint of PQ.

2

(iii) Show that the locus of M as P and Q vary is a branch of a hyperbola.

3

(iv) Show that the equation of the tangent at P is $x + t^2y = 2ct$.

3

(v) The tangent at P meets the x-axis at A and the y-axis at B. Prove that the area of $\triangle OAB$ is constant.

3

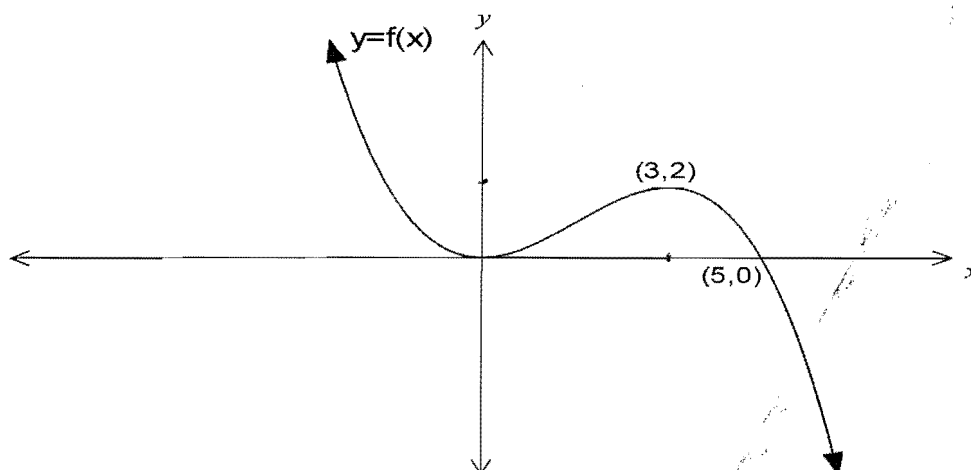
b) Find the value of k if $y = e^{kx}$ is a solution of $y'' - y' - 20y = 0$.

3

Question 5

- a) A relation is defined by $x^2 + xy - 2y^2 = 0$. Show that its gradient has only two values, 1 and $-\frac{1}{2}$. Hence or otherwise, sketch the relation. 3

- b) The diagram shows $y = f(x)$



Draw neat large sketches of

- | | |
|---------------------------|---|
| (i) $y = \frac{1}{f(x)}$ | 3 |
| (ii) $y = [f(x)]^2$ | 2 |
| (iii) $y = f(x) + f(x) $ | 2 |
| (iv) $y = f(x)$ | 2 |
| (v) $y = e^{f(x)}$ | 3 |

Question 6

- a) Draw the graph of $y = |2x - 1| + |x - 2|$ for the domain $-5 \leq x \leq 5$.

Hence or otherwise, solve $|2x - 1| + |x - 2| > 6$. 4

- b) Determine the maximum and minimum values of $\frac{1}{a + b \cos \theta}$ if $a > b > 0$. 3

Question 6 cont

c) For the function $f(x) = \frac{x-2}{x^2-1}$:

(i) Determine the x-values of any stationary points and their nature.

(You do not need to find the y-values).

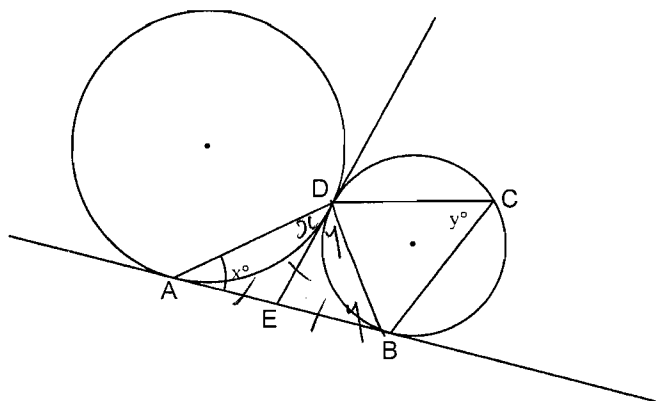
(ii) Sketch $y = f(x)$ showing the important features.

Question 7

a) If $\bar{z}_1 + \bar{z}_2 = 5 + 2i$, find $z_1 + z_2$

b) AB and ED are common tangents to the circles shown.

Prove that the angles marked x and y are complementary



c) (i) If $\frac{p}{q} = \frac{r}{s}$ and $p = rk$, prove that $\frac{p+q}{p-q} = \frac{r+s}{r-s}$.

(ii) If $\frac{p}{q} = \frac{\sin \alpha + \cos \alpha}{\sin \alpha - \cos \alpha}$, simplify $\frac{p+q}{p-q}$.

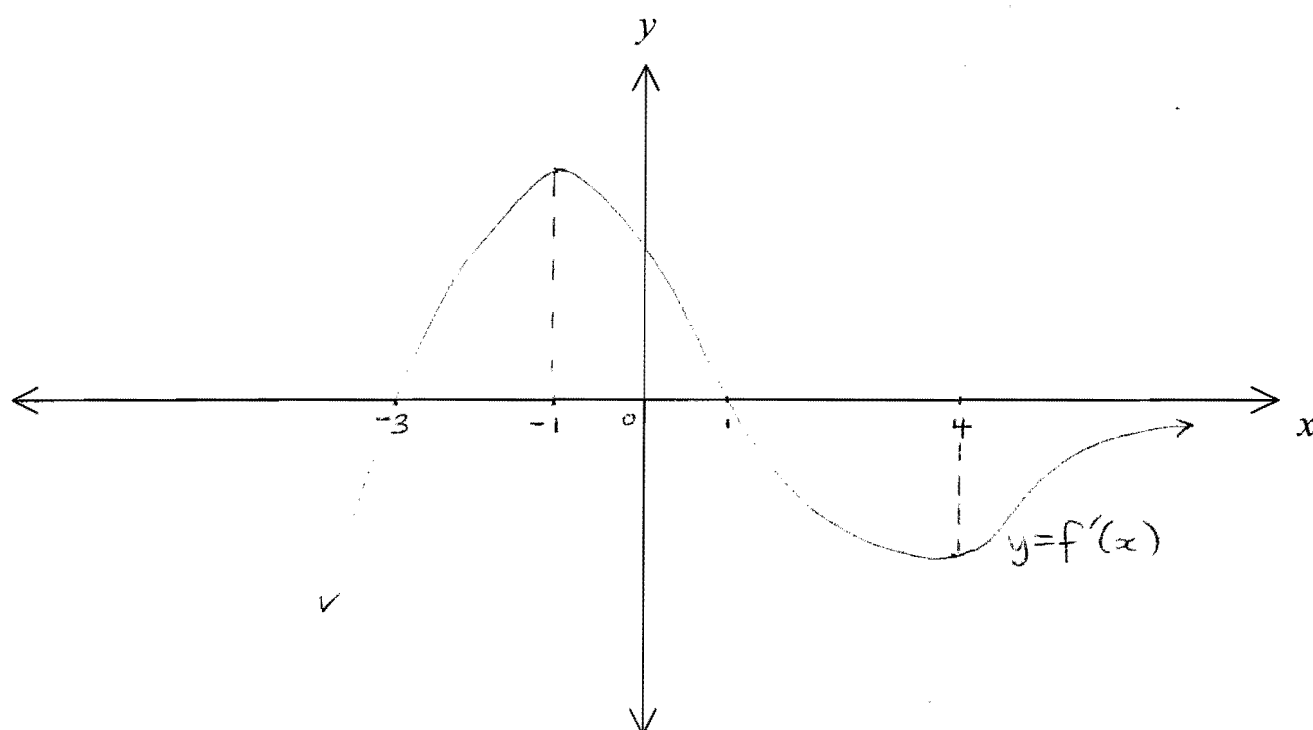
Question 7 (cont)

d) The graph below shows $y = f'(x)$

$$f'(-3) = f'(1) = 0.$$

$$\text{As } x \rightarrow -\infty, f'(x) \rightarrow -\infty.$$

$$\text{As } x \rightarrow \infty, f'(x) \rightarrow 0$$



Sketch the curve $y = f(x)$ given that $f(0)=0$, $f(-3)=-4$, $f(1)=2$ and $f(5)<0$. Clearly show the important features.

Question 8

a) PQ is a chord on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The tangents at P and Q intersect at $T(x_0, y_0)$.

PQ meets a directrix at K.

- (i) Draw a diagram showing all the above information 1
- (ii) Write down the equation of the chord of contact from T. *No proof is required.* 1
- (iii) Find the coordinates of K 2
- (iv) S is the focus which is nearer to K. Show that $\angle TSK = 90^\circ$. 3

b) A and B are angles of a triangle. If $\sin(A-B) = \frac{3}{5}$ and $\sin A \cos B = \frac{4}{5}$,

(i) find the exact value of:

- I. $\cos A \sin B$ 2
- II. $\sin(A+B)$ 1
- III. $\frac{\tan A}{\tan B}$ 2

(ii). Deduce the value of $\tan B$ from (II) and (III) above 3

End of exam

Q1.

$$a) (i) z = \frac{2+i}{2-i} \cdot \frac{2+i}{2+i}$$

$$= \frac{3+4i}{5}$$

$$= \frac{3}{5} + \frac{4}{5}i$$

①

$$(ii) |z| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2}$$

$$= 1$$

①

$$(iii) \arg z = \tan^{-1}\left(\frac{4/5}{3/5}\right)$$

$$= 53^\circ$$

①

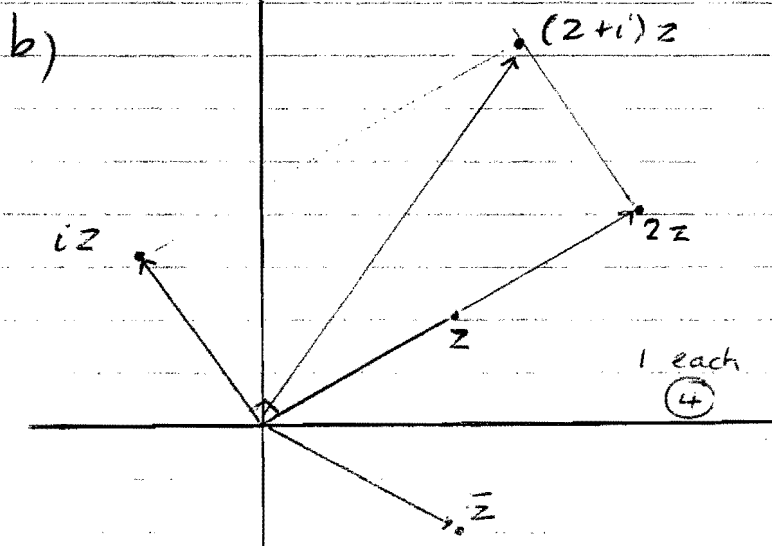
$$(iv) \bar{z} = \frac{3}{5} - \frac{4}{5}i$$

①

$$(v) iz = \frac{3}{5}i - \frac{4}{5}$$

$$\therefore \operatorname{Im}(iz) = \frac{3}{5}$$

①



$$c) (x+iy)^2 = 4-3i$$

$$x^2 - y^2 + 2ixy = 4-3i$$

$$x^2 - y^2 = 4$$

$$2xy = -3$$

$$y = \frac{-3}{2x}$$

$$x^2 - \frac{9}{4x^2} = 4$$

$$4x^4 - 16x^2 - 9 = 0$$

By quad. formula

$$x^2 = \frac{9}{2} \pm \frac{1}{2}$$

$$x = \pm \frac{3}{\sqrt{2}}$$

$$y = \pm \frac{1}{\sqrt{2}}$$

$\therefore$ Roots are $z = \frac{3}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i$

$$\text{and } z = -\frac{3}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i$$

$$d) w = \sqrt{3} + i$$

mod. ①

$$= 2 \operatorname{cis} \frac{\pi}{6}$$

arg ①

$$w^6 = 2^6 \operatorname{cis}\left(6 \times \frac{\pi}{6}\right)$$

$$= 64 \operatorname{cis} \pi$$

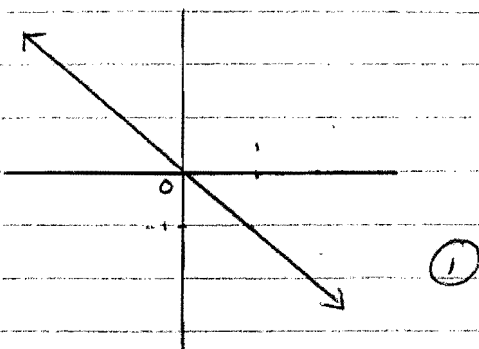
$$= 64(-1 + 0i)$$

$$= -64$$

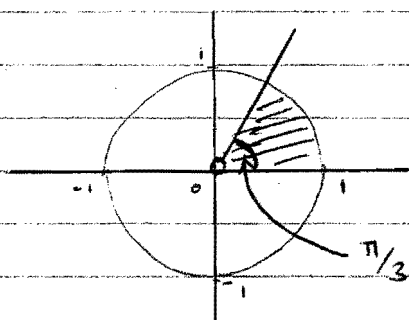
①

Q2

a) (i)



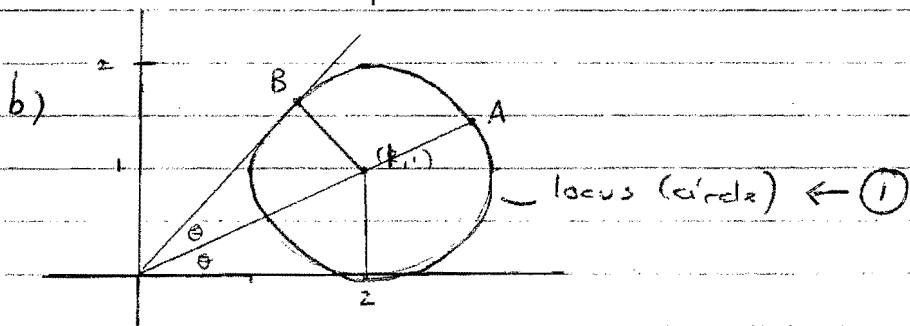
(ii)



① circle

① ray

① shading + empty circle at 0.



Max $|z| = \sqrt{5} + 1$ (at A) ①

Max $\arg z = 2\theta$ (at B)

$= 2 \tan^{-1} \frac{1}{2}$

$= 53^\circ 2'$

①

3
long
retail

e) $(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos^2 \theta \cdot i \sin \theta + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$

$\cos 3\theta + i \sin 3\theta = \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$ ①

$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ (equating reals)

$\cos^3 \theta = \cos 3\theta + 3 \cos \theta \sin^2 \theta$ ①

$\int \cos^3 \theta \, d\theta = \int \cos 3\theta + 3 \cos \theta \sin^2 \theta \, d\theta$

$= \frac{1}{3} \sin 3\theta + \sin^3 \theta + c$

①

①

$\frac{1}{4} [\sin 3\theta + 3 \sin \theta]$

$$d) z_1 z_2 = r_1 (\cos \theta_1 + i \sin \theta_1) \times r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$= r_1 r_2 (\cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 + i^2 \sin \theta_1 \sin \theta_2) \quad (1)$$

$$= r_1 r_2 \left((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i (\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2) \right) \quad (2)$$

$$= r_1 r_2 (\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)) \quad (3)$$

$$\therefore \arg z_1 z_2 = \theta_1 + \theta_2 \quad (4)$$

$$= \arg z_1 + \arg z_2 \quad (5)$$

Q3.

a) $x^2 - y^2 = 0$
 $2x - 2y \cdot \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{-2x}{-2y} = \frac{x}{y}$$

$$\frac{d^2y}{dx^2} = \frac{y \cdot 1 - x \cdot \frac{dy}{dx}}{y^2}$$

$$= \frac{y - x \cdot \frac{x}{y}}{y^2}$$

$$= \frac{\frac{y^2}{y} - \frac{x^2}{y}}{y^2}$$

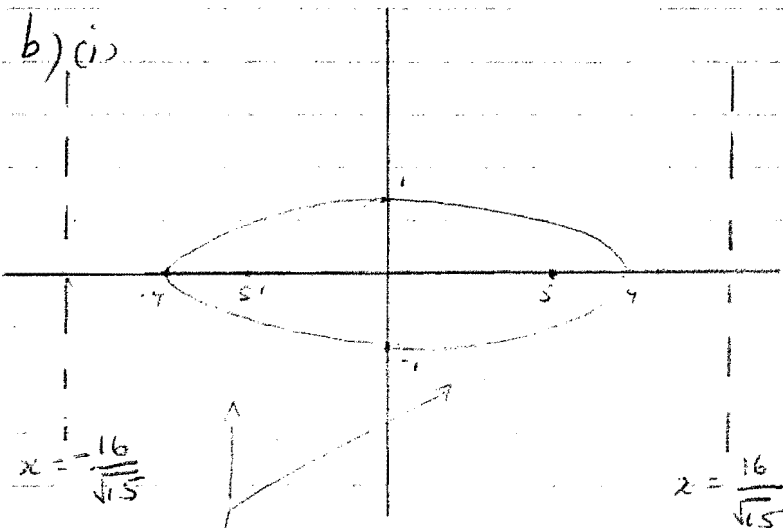
$$= \frac{y^2 - x^2}{y^3}$$

$$= \frac{0}{y^3}$$

$$= 0$$

but
 $x^2 - y^2 = 0$
 $y^2 - x^2 = 0$

b) (i)



$$S(\sqrt{5}, 0)$$

$$S'(-\sqrt{5}, 0)$$

x -int ①
 y -int ①
 Foc ①
 Dir ①

(ii) $x = 4 \cos \theta$ ①

$y = \sin \theta$ ①

(iii) When $x = 2$, $\frac{4}{16} + y^2 = 1$

$$y^2 = \frac{12}{16}$$

$$y = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2} \quad (y > 0)$$

So pt of contact is $(2, \frac{\sqrt{3}}{2})$ ①

$$\frac{2x}{16} + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x/8}{2y}$$

$$= -\frac{x}{16y}$$

$$= -\frac{2}{16 \cdot \frac{\sqrt{3}}{2}}$$

$$= -\frac{2}{8\sqrt{3}}$$

$$= -\frac{1}{4\sqrt{3}}$$

①

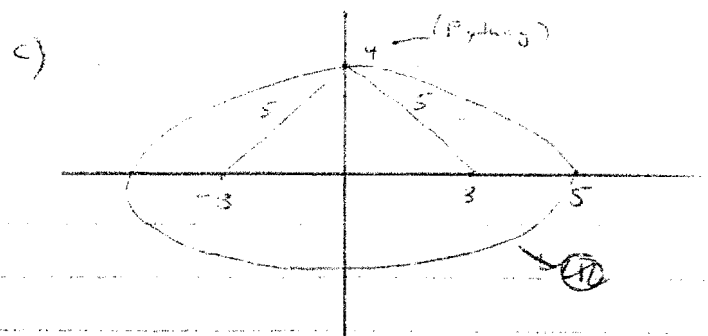
$$y - \frac{\sqrt{3}}{2} = -\frac{1}{4\sqrt{3}}(x - 2)$$

$$4\sqrt{3}(y - \frac{\sqrt{3}}{2}) = -x + 2$$

$$4\sqrt{3}y - 6 = -x + 2$$

$$x + 4\sqrt{3}y - 8 = 0$$

②

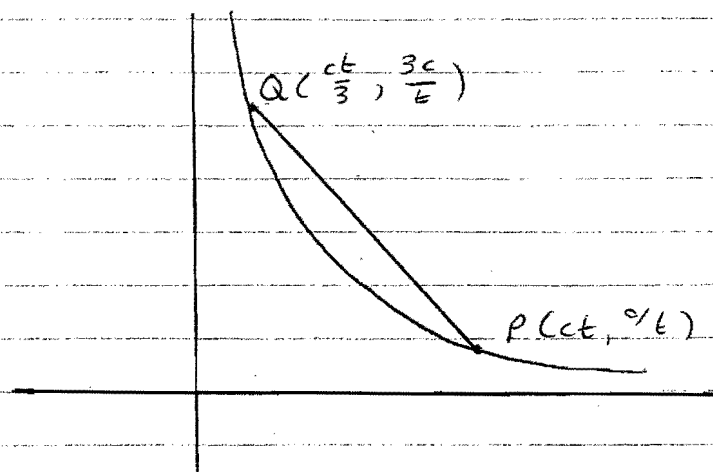


$$\frac{x^2}{25} + \frac{y^2}{16} = 1 \quad \text{An ellipse ①}$$

$$\text{Eqn ②}$$

Q4.

a) (i)



Note: Q is to left of P.

$$(ii) M = \left(\frac{ct + \frac{ct}{3}}{2}, \frac{\frac{3c}{t} + \frac{c}{t}}{2} \right)$$

$$= \left(\frac{\frac{4ct}{3}}{2}, \frac{\frac{4c}{t}}{2} \right)$$

$$= \left(\frac{2ct}{3}, \frac{2c}{t} \right)$$

① x-value

① y-value

$$(iii) \text{ Since } x = \frac{2ct}{3}, y = \frac{2c}{t}$$

$$xy = \frac{2ct}{3} \cdot \frac{2c}{t} = \frac{4c^2}{3} \text{ ① which is constant.}$$

Since $xy = \text{constant}$ ① locus is a hyperbola. Since $t > 0$, only ~~the~~ one branch. ①

(iv) Tangent

$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$\left(y = \frac{c^2}{x} \right)$$

$$= -\frac{c^2}{x^2} \text{ ①}$$

$$= \frac{-c^2}{(ct)^2}$$

at P (ct, c/t)

$$= \frac{-c^2}{c^2 t^2}$$

$$= -\frac{1}{t^2} \text{ ①}$$

$$y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

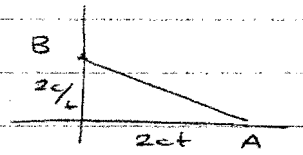
$$t^2 (y - \frac{c}{t}) = -x + ct \text{ ①}$$

$$t^2 y - ct = -x + ct$$

(v) Tang. at P $x + t^2 y = 2ct$

At A: $y=0$ $x=2ct$ $A(2ct, 0)$ ①

At B: $x=0$ $t^2 y = 2ct$
 $y = \frac{2c}{t}$ $B(0, \frac{2c}{t})$ ①



$$\text{Area} = \frac{1}{2}bh = \frac{1}{2} \cdot 2ct \cdot \frac{2c}{t}$$

$$= 2c^2 \text{ which is constant.} \quad \textcircled{1}$$

b) $y = e^{kx}$
 $y' = k e^{kx}$
 $y'' = k^2 e^{kx}$ } ①

$\therefore y'' - y' - 20y = (k^2 - k - 1)e^{kx} = 0$ ① only when

$$k^2 - k - 1 = 0$$

$$k = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2} \quad \textcircled{1}$$

Question 5.

a) $(x+2y)(x-y)=0$

$$x = -2y$$

$$y = -\frac{1}{2}x$$

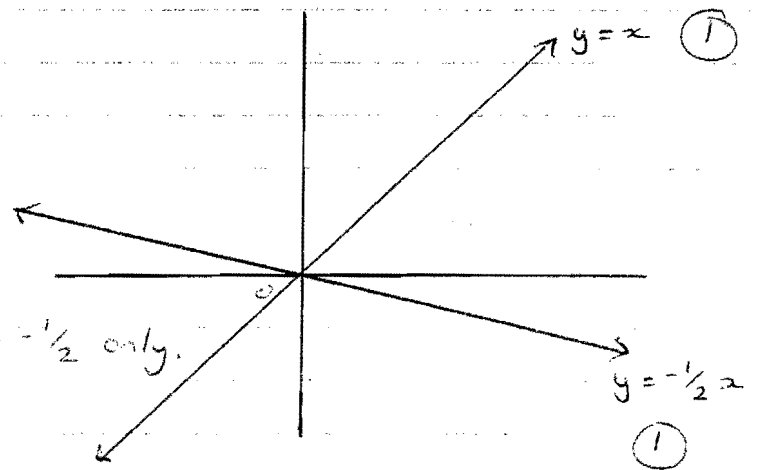
$$(m = -\frac{1}{2})$$

$$x = y$$

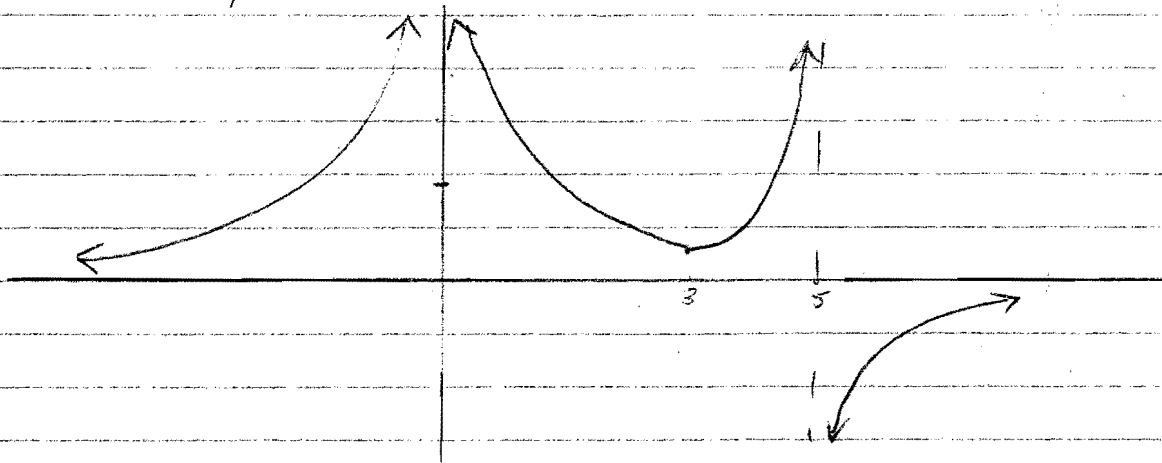
$$(m = 1)$$

$$\textcircled{1}$$

So $m = \frac{dy}{dx} = 1 \text{ or } -\frac{1}{2} \text{ only.}$



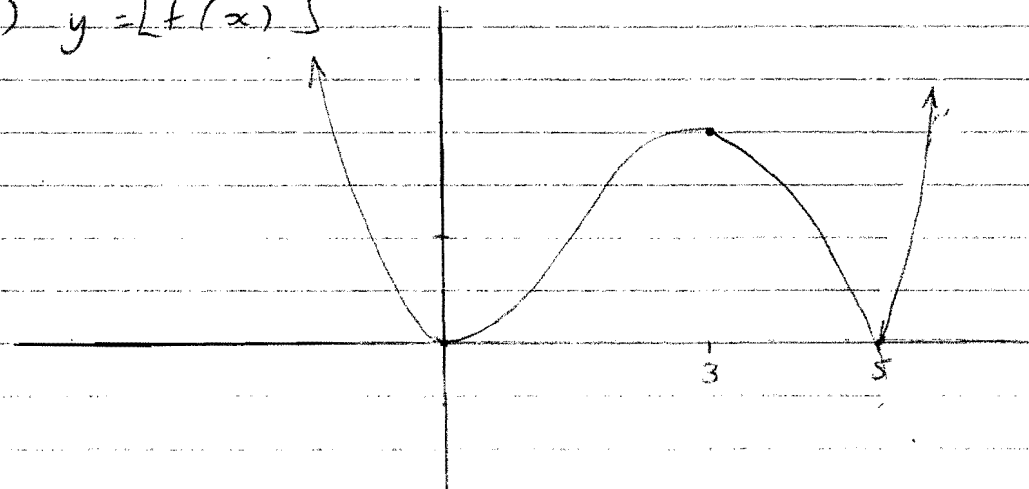
(i) $y = \frac{1}{f(x)}$



③

1 each section

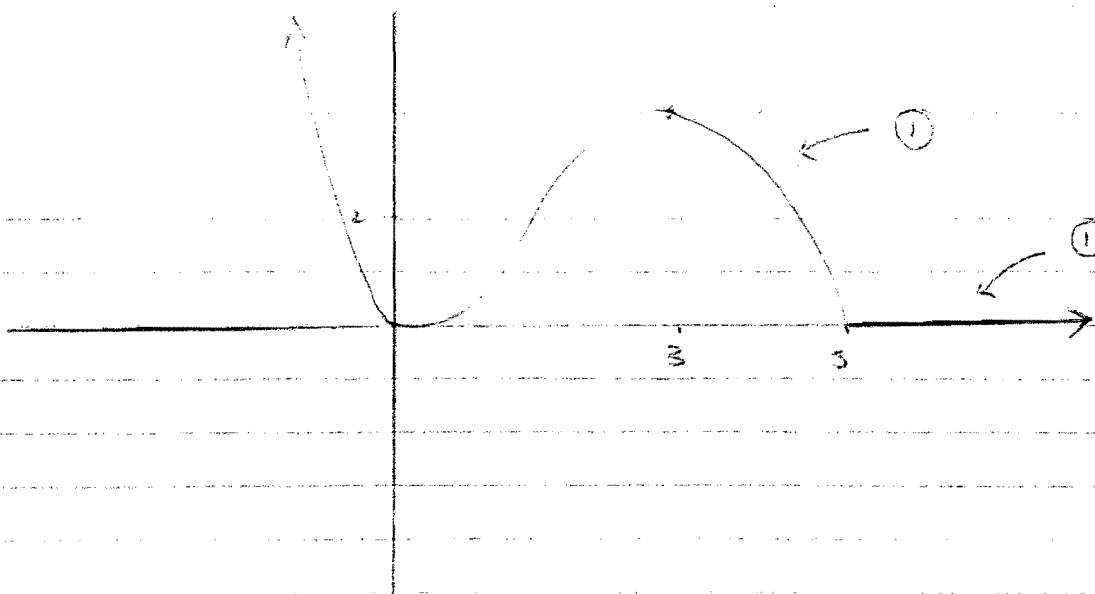
(ii) $y = [f(x)]^2$



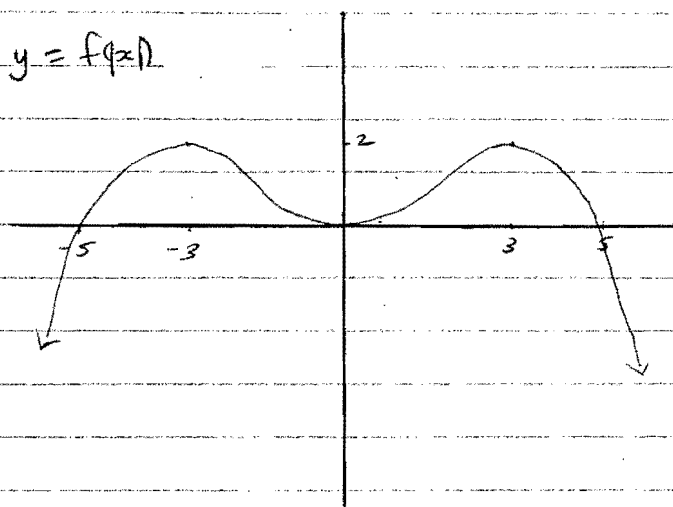
②

(iii) $y = f(x) + |f(x)|$

$= 0$ when $f(x)$ neg
 $= 2f(x)$ when $f(x)$ pos

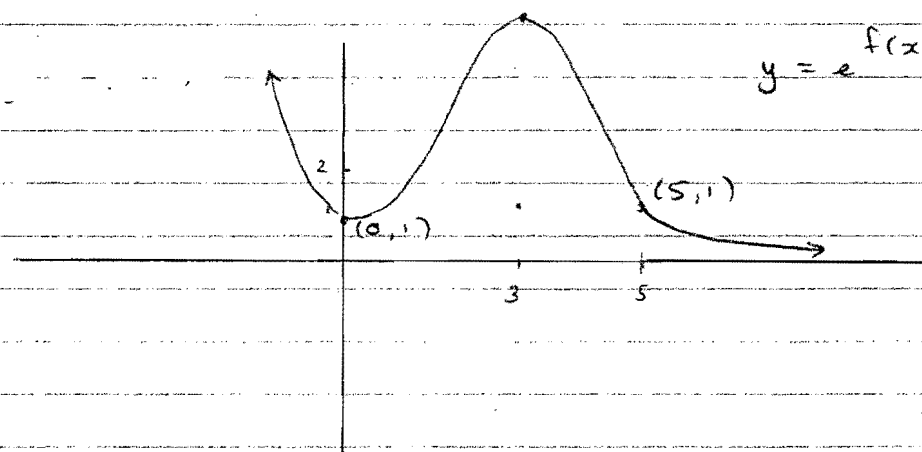


(iv) $y = f(x)$



(2)

(v)

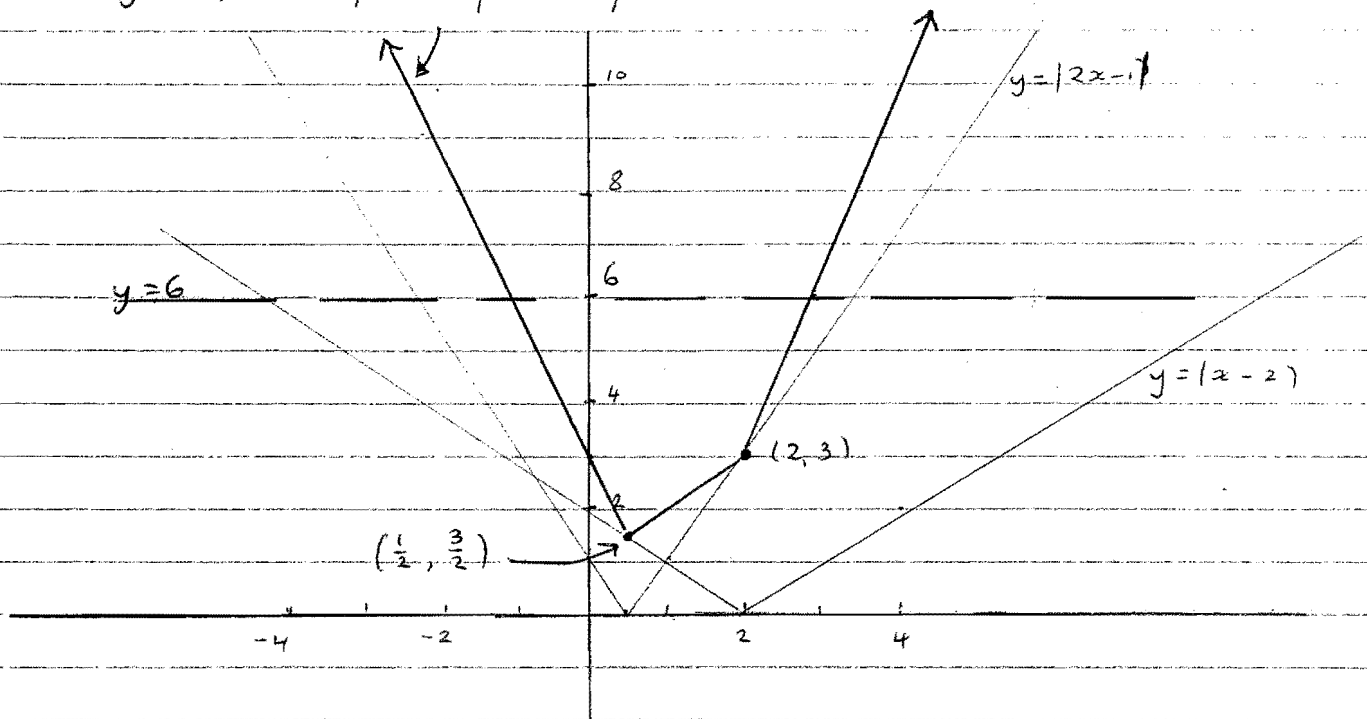


$y = f(x)$

(3)

Q6.

a) $y = |2x-1| + |x-2|$



$y=6$ intersects when $x=-1, x=3$ (Can solve eqns or check by substitution)

Soln to inequality: $x \leq -1$, $x \geq 3$.

b) For $a + b \cos \theta$, $\max \text{ value} = a+b$ } and both results are positive
 $\min \text{ value} = a-b$
 For $\frac{1}{a+b \cos \theta}$, $\max \text{ value} = \frac{1}{a-b}$
 $\min \text{ value} = \frac{1}{a+b}$

c) $f(x) = \frac{x-2}{x^2-1} = \frac{x-2}{(x-1)(x+1)}$

(1) $f'(x) = \frac{(x^2-1) \cdot 1 - (x-2) \cdot 2x}{(x^2-1)^2} = \frac{x^2-1-2x^2+4x}{(x^2-1)^2}$

$= \frac{-x^2+4x-1}{(x^2-1)^2}$

For stat. pts $f'(x) = 0$: when

$x^2-4x+1=0$
 $x = \frac{4 \pm \sqrt{16-4}}{2}$
 $= \frac{4 \pm 2\sqrt{3}}{2}$

$$= 2 + \sqrt{3}$$

Use the first deriv. to determine nature

When $x = 2 + \sqrt{3} \approx 3.7$

x	3	$2 + \sqrt{3}$	4	
$f'(x)$	+	0	-	$\swarrow$
	$-9+12-1$		$-16+16-1$	

[Sign of $f'(x)$ = Sign of $-x^2 + 4x - 1$] Max turn pt
when $x = 2 + \sqrt{3}$

When $x = 2 - \sqrt{3} \approx 0.3$

x	0	$2 - \sqrt{3}$	1	
$f'(x)$	-	0	+	$\swarrow$
	$-0+0-1$		$-1+4-1$	

Min. turn pt when $x = 2 - \sqrt{3}$

(ii) $f(x) = \frac{x-2}{x^2-1} = \frac{x-2}{(x-1)(x+1)}$

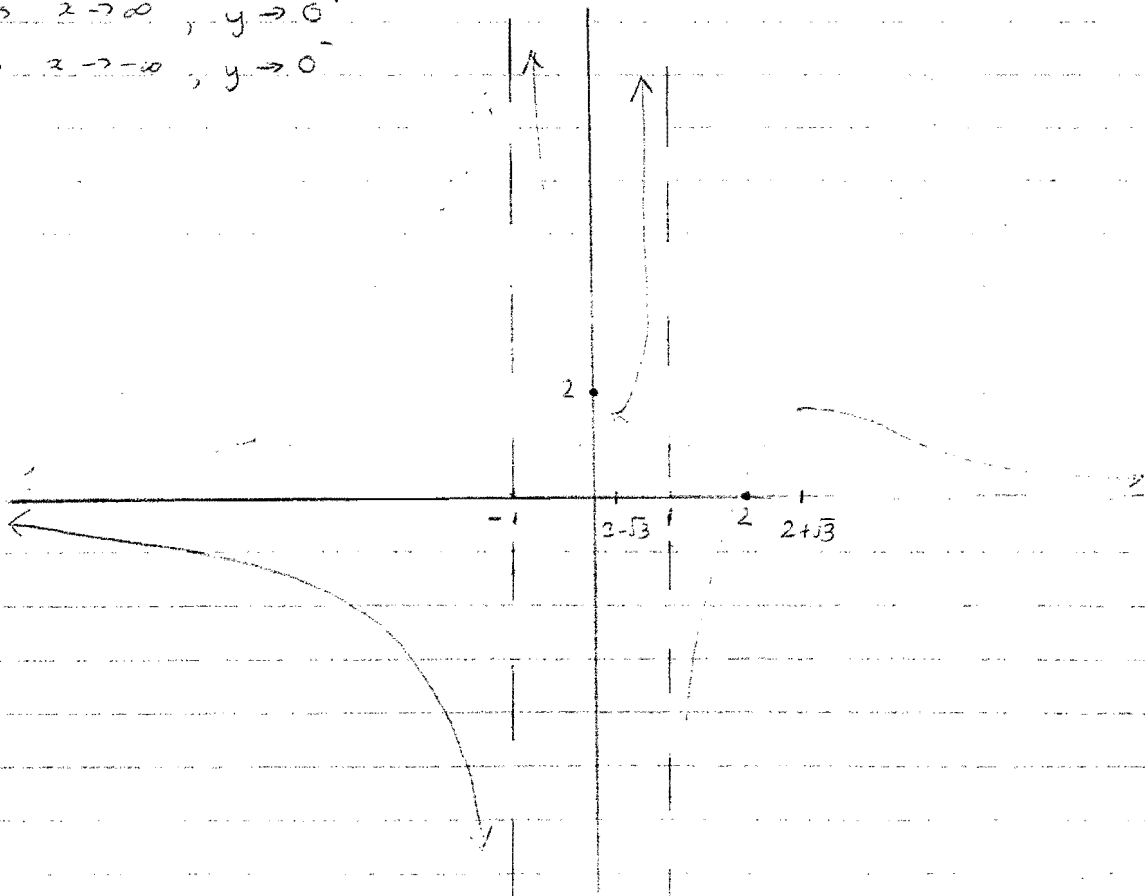
Vert asymp. $x = 1, -1$

Horiz. asymp. $y = 0$

Start pts as above

As $x \rightarrow \infty, y \rightarrow 0^+$

As $x \rightarrow -\infty, y \rightarrow 0^-$



Q7.

a) $S - 2i$

b) Join DB. Draw circle with centre E; through A, D, B. ①

$$\angle ADE = x \quad (\text{base } \angle s, \text{ isos } \triangle) \quad ①$$

$$\angle DEB = 2x \quad (\text{ext } \angle \text{ of } \triangle = \text{sum of int. opp } \angle s) \quad ①$$

$$\angle EBD = \angle EPB \quad (\text{base } \angle s, \text{ isos } \triangle)$$

$$= \frac{180 - 2x}{2} \quad (\angle \text{ sum of } \triangle)$$

$$= 90 - x \quad ①$$

$$y = 90 - x \quad (\angle \text{ between tangent + chord} \\ = \angle \text{ in alt. seg.}) \quad ①$$

$$x + y = 90$$

c) (i) $\frac{rk}{q} = \frac{r}{s} \quad (\text{since } p=rk)$

$$\frac{k}{q} = \frac{1}{s}$$

$$\underline{q = sk}$$

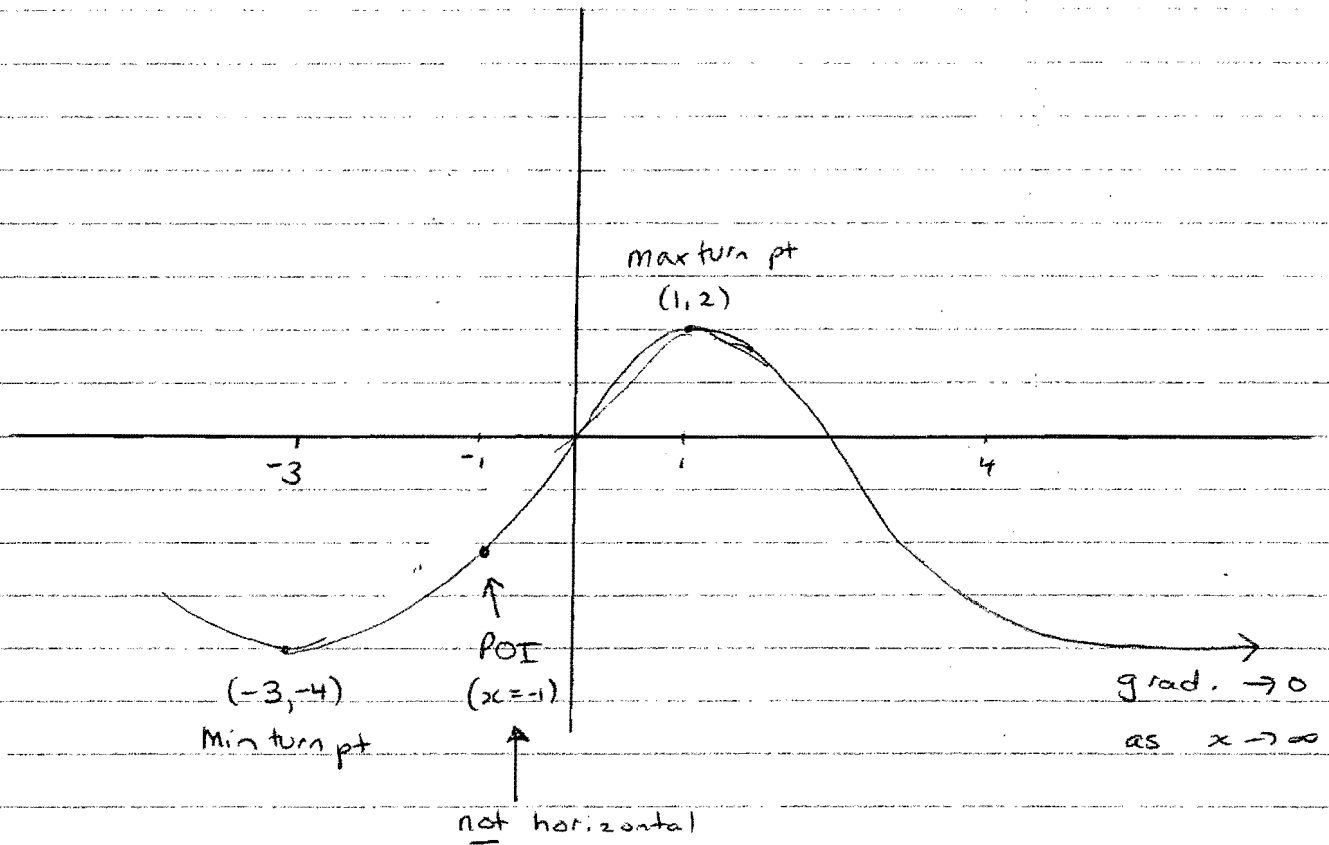
$$\frac{p+q}{p-q} = \frac{rk+sk}{rk-sk} = \frac{k(r+s)}{k(r-s)} \\ = \frac{r+s}{r-s}$$

(ii) If $\frac{p}{q} = \frac{\sin \alpha + \cos \alpha}{\sin \alpha - \cos \alpha} \quad \leftarrow kr \text{ as } \leftarrow ks$

then $\frac{p+q}{p-q} = \frac{kr+ks}{kr-ks}$

$$= \frac{\sin \alpha + \cos \alpha}{\sin \alpha - \cos \alpha} \div \frac{\sin \alpha + \cos \alpha}{\sin \alpha - \cos \alpha} \\ = \frac{2 \sin \alpha}{2 \cos \alpha} \\ = \tan \alpha$$

d)



Thr (0, 0) ①

Min (-3, -4) ①

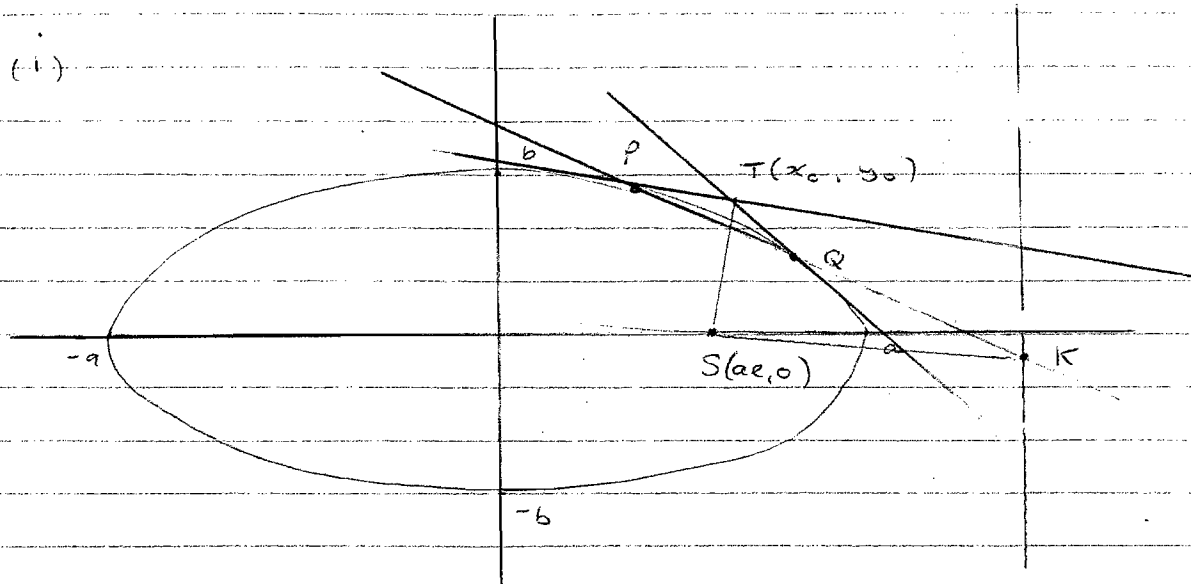
Max (1, 2) ①

POI, not horiz ①

y neg + m $\rightarrow 0$ as $x \rightarrow \infty$ ①

Q8.

a) (i)



$$(i) \quad \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$$

$$(ii) \quad \text{At } K: \quad x = \frac{a}{e}$$

$$\frac{\frac{a}{e} \cdot x_0}{a^2} + \frac{yy_0}{b^2} = 1$$

$$\frac{x_0}{ae} + \frac{yy_0}{b^2} = 1$$

$$y = \left(1 - \frac{x_0}{ae}\right) \cdot \frac{b^2}{y_0}$$

$$K \left(\frac{a}{e}, \frac{b^2}{y_0} \left(1 - \frac{x_0}{ae}\right) \right)$$

$$(iv) \quad m_{TS} = \frac{y_0 - 0}{x_0 - ae} = \frac{y_0}{x_0 - ae}$$

$$m_{SK} = \frac{\frac{b^2}{y_0} \left(1 - \frac{x_0}{ae}\right) - 0}{\frac{a}{e} - ae}$$

$$= \frac{\frac{b^2}{y_0} \cdot \frac{ae - x_0}{ae}}{\frac{a - ae^2}{e}} \times \frac{e}{e}$$

$$= \frac{\frac{b^2}{ay_0} (ae - x_0)}{a(1 - e^2)}$$

$$\text{but } b^2 = a^2(1 - e^2)$$

$$= \frac{b^2 (ae - x_0)}{a^2 y_0 (1 - e^2)}$$

$$= \frac{ae - x_0}{y_0}$$

$$m_{TS} \times m_{Sk} = \frac{y_0}{x_0 - ae} \times \frac{ae - x_0}{y_0} = -1$$

$$\therefore \angle TSk = 90^\circ$$

b) (i) (I) $\sin(A-B) = \underbrace{\sin A \cos B}_{\frac{4}{5} \text{ given}} - \underbrace{\cos A \sin B}_{\frac{1}{5} \text{ follows}} = \frac{3}{5}$ / expansion given

(II) $\cos A \sin B = \frac{1}{5}$ /

(II) $\sin(A+B) = \sin A \cos B + \cos A \sin B$
 $= \frac{4}{5} + \frac{1}{5}$
 $= 1$ /

(III) $\frac{\tan A}{\tan B} = \frac{\sin A}{\cos A} \div \frac{\sin B}{\cos B}$
 $= \frac{\sin A \cos B}{\cos A \sin B}$
 $= \frac{4/5}{1/5}$
 $= 4$ /

(ii) $\frac{\tan A}{\tan B} = 4$

$$\tan B = \frac{1}{4} \tan A$$

$\sin(A+B) = 1$ and A, B are $\angle$ s of Δ

$A+B = 90^\circ$, ~~or~~ only /

$A = 90^\circ - B$

$\tan B = \frac{1}{4} \tan(90^\circ - B)$

$\tan B = \frac{1}{4} \cot B$ using complementary ratios

$\tan B = \frac{1}{4 \cot B}$ using reciprocal ratios

$$\tan^2 B = \frac{1}{4}$$

$$\tan B = \pm \frac{1}{2}$$

but

since $A+B=90^\circ$

and A, B are $\angle$ s of $\triangle$

$$B \sim 30^\circ, 150^\circ$$

$$\tan B = \frac{1}{2} \text{ only.}$$